

Smart Building Analytics & Energy Digital Twin - FY2025 Report

Keppel Bay Tower · Office · Singapore

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1. Background

Before 2023, operational data at Keppel Bay Tower was spread across 17 separate data silos: the BMS, lighting controls, lift monitoring, power meters, the car park system and several vendor portals. Each system used its own naming conventions, so combining data for analysis took days of manual work and nobody had a complete, real-time view of how the building used energy.

2. Integrated smart-building platform

A cloud IoT platform was deployed to connect 170 devices, 30 systems and 65 sub-systems, collecting data from more than 20,000 sources. The key design decision was a standard point-naming and tagging scheme, inspired by the Brick Schema, applied to every sensor and meter. With standard naming, an analytic written once - for example 'detect a chiller running with no load' - can be reused on any building that follows the same scheme.

- Real-time dashboards for energy use intensity, plant efficiency and indoor environment.
- Automated fault detection, with work orders created in the maintenance system (CMMS).
- Tenant-level metering of after-hours air-conditioning use.

Note: the numbers of data silos, devices and data sources follow publicly reported descriptions of the Keppel Bay Tower smart-building programme; all other figures in this corpus are synthetic.

3. Energy digital twin

A physics-based energy model of the building was calibrated against twelve months of metered data until its predictions matched measurements within ASHRAE Guideline 14 tolerances, with a monthly error below 1%. The digital twin was then used to test energy conservation measures (ECMs) virtually, before any change was made to the real building.

Energy conservation measure	Predicted saving	Status
AHU schedules aligned to actual occupancy	1.6%	Implemented
Perimeter-zone daylight dimming	1.1%	Implemented
Chilled-water reset tuning	1.2%	Implemented
Car park ventilation controlled by CO sensors	0.9%	Implemented
Tenant after-hours AC tracking and billing	0.7%	Implemented
Condenser pump setpoint tuning	0.6%	Implemented
Toilet exhaust fan scheduling	0.5%	Implemented
Lift standby mode in off-peak hours	0.4%	Implemented

All eight ECMs were implemented within three months. Measured energy use intensity fell by 7%, in line with the digital-twin prediction.

4. Lessons learned

- Data governance comes first. Agreeing a single naming standard took longer than installing the platform, but it is what makes analytics reusable across buildings.
- Test before touching. The digital twin rejected two proposed measures that would have caused comfort problems in west-facing zones.
- Recalibrate after major changes. The twin drifted after the chiller plant retrofit and had to be recalibrated with the new plant performance curves.
- Close the loop to maintenance. Analytics only saved energy once alerts were routed automatically into CMMS work orders with an owner and a deadline.

5. Scaling across the portfolio

The naming standard, dashboards and fault-detection rules are asset-agnostic. The same data model can sit on top of a data centre DCIM or the BMS of a residential development, so that energy analytics, anomaly detection and benchmarking are written once and reused everywhere.